# claude-windows / 985cc286-9e1e-4217-87c3-2d418bc4fa9c

## Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\985cc286-9e1e-4217-87c3-2d418bc4fa9c\subagents\agent-a4c1b6c682a81ff69.jsonl`
- SHA-256: `5ad6ac88327f5c53659ffa91a2dc943d427843dfa5ffde9e05006fe03a13a2bb`
- Source modified: `2026-06-14T09:53:32+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `subagents`
- session_id: `985cc286-9e1e-4217-87c3-2d418bc4fa9c`

## Transcript

### 1. user (2026-06-14T09:49:18.419Z)

Quality checkpoint: update the iteration docs in the badminton-highlight-indexer repo (cwd C:/Users/avidu/Projects/badminton-highlight-indexer). Work in an ISOLATED worktree off origin/master: `git -C C:/Users/avidu/Projects/badminton-highlight-indexer fetch origin` then `git -C ... worktree add C:/Users/avidu/Projects/bhi-qiter -b docs/quality-checkpoint-personalization origin/master`.

(1) **docs/QUALITY_ITERATIONS.md** ? add a NEW entry at the TOP of the "## Current track (live) ? newest first" section, matching the existing `### Title *(date)*` style. Capture the **PERSONALIZATION Phase-0 quality checkpoint**: the owner adopted the HYBRID (personalization as a flag-gated FEATURE on a generalization-first baseline + a contribution flywheel ? full design in `docs/PERSONALIZATION_PLAN.md`, merged #144), and the three honest-eval RAILS landed this checkpoint ? **per-user promotion gate (#141:** min-N floor + structural-guard exemption ? per-user/small-N tuning can't promote noise or disable a structural guard), **held-out-USER learning curve (#142:** turns "<K videos to superior quality" into a falsifiable release number), **per-user no-regression guard (#143:** a retrain/upgrade can never make a user worse than their incumbent). Frame HONESTLY: these are the eval/safety RAILS for the personalization feature, NOT new measured rally-detection wins. Note the locked owner decisions (identity = north-star-design/current-impl; inline classifier behind a cloud-load interface; **`min_n` is a promotion-confidence floor, NOT a service gate**; free-tier video pooling; **Gate-A-only structural + `min_n=5` now LOCKED**).

(2) **docs/PERSONALIZATION_PLAN.md** ? in the "§1 / Pending sub-decision" area, change the structural-guard line from "(Awaiting owner confirmation.)" to LOCKED: "**Gate-A-only structural, `min_n = 5` ? owner-confirmed 2026-06-14.**"

Branch-safe; commit explicit paths (end msg with `Co-Authored-By: Claude Opus 4.8 (1M context) <noreply@anthropic.com>`); push; `gh pr create --base master`; then MERGE on green CI (docs/quality-log update, within the merge agreement): `gh pr merge --squash --delete-branch`, `git worktree remove C:/Users/avidu/Projects/bhi-qiter --force`, `git branch -D docs/quality-checkpoint-personalization`, `git worktree prune`. If CI is still pending after a reasonable wait, push the merge anyway is NOT allowed ? instead report the PR number as OPEN+green-pending for the parent to merge. Return: PR number/URL + merge status.

### 2. assistant (2026-06-14T09:49:24.699Z)

I'll work through this quality checkpoint task. Let me start by setting up the isolated worktree off origin/master.

### 3. user (2026-06-14T09:49:27.104Z)

(Bash completed with no output)

### 4. assistant (2026-06-14T09:49:29.062Z)

Now let me create the isolated worktree.

### 5. user (2026-06-14T09:49:30.745Z)

```
Preparing worktree (new branch 'docs/quality-checkpoint-personalization')
branch 'docs/quality-checkpoint-personalization' set up to track 'origin/master'.
HEAD is now at 4be16e1 docs(personalization): add PERSONALIZATION_PLAN ? hybrid per-user feature + locked owner decisions (#144)
```

### 6. assistant (2026-06-14T09:49:35.984Z)

The worktree is set up. Note the HEAD is at #144 (the personalization plan merge). Let me read both target files to understand the existing structure.

### 7. user (2026-06-14T09:49:36.613Z)

```
# Rally-detection quality iterations ? living log

**Purpose.** A running, reverse-chronological log of how we evolve rally-detection quality: each
iteration's hypothesis, what shipped, the measured (or expected) effect, and the lesson. This is the
**live** continuation of the historical trail in
[`archives/quality-iterations/`](archives/quality-iterations/README.md) ? per the archive policy
(`CLAUDE.md`), **terminal-state** (DONE/REJECTED) work gets a dated subdir there; this file stays at
`docs/` top level and keeps growing, then its entries graduate to the archive when they conclude.

**Methodology (non-negotiable).** Every quality change rides the experiment harness
([EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md)): new cues land **flag-gated, default OFF**; they're
graded **offline against golden labels** (per-video + LOVO-honest); promotion to default happens **only
on measured lift** through the `run_regression.py` gate; **rollback = flip a config flag, never
`git revert`.** Cues are designed in [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md); scoring
in [EVALUATION.md](EVALUATION.md); the 2-layer model in
[HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md).

**Golden corpus = the 6 labelled matches** in `output/human_lovo_manifest.json` (owner-confirmed
2026-06-13: use all 6 ? `adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy,
testlarge_short, mahadevpura_singles`). The model trained on these is what we run alongside the existing
detector on NEW footage for the side-by-side review.

---

## Baselines that drive strategy (IoU 0.5 vs human golden labels)

| Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
|---|---|---|
| Heuristic (velocity windowing) | 0.657 | 0.157 |
| Gen-0 owned head (LOVO, n=6) | 0.744 | 0.561 |
| Two-stage WASB+Gemini refine | 0.83 | ? |
| Gemini wrapper (`gemini-flash-latest`) | 0.93 | 0.66 |

**Reading:** clean footage is commoditized (serve via Gemini); **multi-court drops the commodity wrapper
to 0.66 ? that is the owned model's ship target.** Baselines tracked in `eval_baselines/`. The owner's
observed false positive ? a player **holding/flicking** the shuttle between points counted as a rally ?
```

is the precision bug the fusion gates target.

---

## Current track (live) ? newest first

### Fusion P3 + P4 MEASURED on the n=6 golden corpus ? person = no lift, audio = promising (not promotable) *(2026-06-14)*
The GPU person-feature extraction finished all 6 golden videos (`fusion_golden.py` ? `output/*_golden_features.json`, the
replayable substrate); the CPU LOVO drivers then measured the lift with **honest stats (C1 ? bootstrap 95% CI + paired sign
test; never a bare mean at n=6)**:

| Variant (honest LOVO, IoU ? 0.5) | F1 | ?F1 vs prior | 95% CI | sign test | CI clears 0 |
|---|---|---|---|---|---|
| shuttle-only | 0.307 | ? | ? | ? | ? |
| + person (P3) | 0.286 | **?0.021** (?6.7%) | [?0.165, +0.162] | 2W/4L (p=0.688) | **No** |
| + person + audio (P4 incr.) | 0.366 | **+0.079** (+27.7%) | [?0.054, +0.203] | 4W/2L (p=0.688) | **No** |

**Read honestly:** neither cue clears the promotion bar (both CIs span 0, both sign tests n.s.) at n=6 ? **both stay
default-OFF** (promote-only-on-lift). **Person** adds noise (no court polygons ? `players_in_court` counts spectators);
per-video it's a wash (testlarge +0.40, gbaaddy ?0.22). **Audio** is the more promising lead ? +0.079 / +27.7%, wins 4/6
(largetest +0.27, mahadevpura +0.27), and shuttle+person+audio (0.366) lands **above** shuttle-only (0.307) ? but n=6 can't
confirm it. **Next:** more matches (raise N until the sign test can resolve ~+0.08), and measure audio on shuttle-only
*directly* (without the person handicap). Recorded in `eval_baselines/experiment_leaderboard.json`. **Negative/uncertain
results are kept on purpose ? this is the honest ablation backbone for the paper aim.**

### Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125 corroboration *(2026-06-14)*
**Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness).** The measured win
(?F1 +0.074, **6/6 golden matches, sign-test p=0.031**, CI [+0.042,+0.104]; over-seg 1.68?1.01 ? the
[first-A/B archive](archives/quality-iterations/2026-06-first-ab-experiment/README.md)) clears the honest
bar. **Recommended config:** `indexing.fusion.min_crossings = 2` on the fusion path (the eval-only Gate A
folded into production, MULTI_SIGNAL_FUSION_PLAN §3.1). ? The offline win is on the recall-oriented *proposal*
candidate set; flipping the **live served default** stays owner-gated until a real fusion-path run confirms it
(the production fusion segmenter is P2/unverified on real video) ? so this records the promotion + rationale,
not a silent default flip. **Corroborated on real footage:** offline re-score of GX010125_1080p's *cached*
trajectory (no GPU, isolated `output/GX010125_run/`) ? 45 candidates ? CONTROL 45 rallies vs **Gate-A 27**;
Gate-A dropped **18 low-crossing windows (~1.7 min)** of likely held-shuttle/feed FPs (no golden labels for
this footage ? divergence, not F1).

**Learned variant FIXED ? [#130](https://github.com/avidullu/badminton-highlight-indexer/pull/130).** The
first A/B's learned logistic lost to CONTROL and zeroed `testlarge` ? a fixed-0.5-cutoff artifact. Added

additive/default-OFF `Variant.tune_threshold` (operating point chosen on the **train** fold) + `calibrate`
(Platt/isotonic, C2). Validated on the n=6 golden set: **mean F1 0.307 ? 0.423 (+0.116)**, testlarge
0.000 ? 0.300, and the learned variant now **beats CONTROL (0.388)**. Threshold-in-LOVO is the big lever;
calibration is available too.

**Next:** a real fusion-path run to confirm Gate-A served-side; per-video output layout
([PER_VIDEO_OUTPUT_LAYOUT.md](PER_VIDEO_OUTPUT_LAYOUT.md)) so each video's artifacts are self-contained.

### First end-to-end A/B on the golden corpus (n=6) ? **ARCHIVED** *(2026-06-14)*
First real use of the harness + C1?C4 helpers, 100% offline (re-scored persisted WASB trajectories, no GPU).
**Gate-A (net-crossings ? 2) is a measured win** ? ?F1 +0.074, 6/6 matches, sign-test **p = 0.031**, CI
[+0.042, +0.104], over-segmentation (segment-count ratio) **1.68 ? 1.01**. The window-level 5-feature logistic
**did not clear the bar** (?0.081, 2W/4L; a threshold/calibration artifact ? *not* the stronger per-frame
prototype). The framework held up: **C1 promoted one variant and refused another; C4 surfaced over-segmentation
tIoU hid.** Full record + tables ?
[`archives/quality-iterations/2026-06-first-ab-experiment/`](archives/quality-iterations/2026-06-first-ab-experiment/README.md).
Next: Gate-A is a candidate for the no-regression promotion gate; the logistic failure is the C2 (calibration)
+ threshold-in-loop task list.

### Eval-honesty helpers (course corrections C1?C4) ? additive, default-OFF *(2026-06-14)*
**Why.** A multi-agent literature review (folded into
[ANALYZERS_AND_RECONCILERS.md](ANALYZERS_AND_RECONCILERS.md) §13) confirmed the framework's *skeleton* is
standard prior art (Temporal Action Localization/Segmentation + late/score-level fusion ? cite, don't claim),
and flagged that our **measurement** is walking into known small-N traps that would make "promote only on
measured lift" promote noise or hide a real win. Four corrections, shipped as **pure, additive helpers** (no
existing `compare`/LOVO/`metrics.score` behaviour changed ? callers opt in):
- **C1 ? never a single mean.** `backend/eval/eval_stats.py`: `mean_with_ci`/`bootstrap_ci` (deterministic,
  seeded), `paired_sign_test` (the exact distribution-free "won on enough *videos*?" test the corpus already
  needs), `paired_delta_ci` (B?A with a CI that must clear 0). At n?6 a lone LOVO mean is not trustworthy.
- **C2 ? calibrate cue confidence.** `backend/eval/score_calibration.py`: Platt + isotonic (PAVA) calibrators
  + `brier_score`/`expected_calibration_error`, so a CNN softmax, an audio onset, and an integer net-crossing
  count are comparable before the arbiter and the over-confident cue can't swamp the rest.
- **C3 ? honest stacking.** `eval_stats.out_of_fold_predictions`/`grouped_folds` (leave-one-group-out by
  match) so a *learned* producer's outputs used as arbiter features are out-of-fold (no leakage).
- **C4 ? over-segmentation-visible metrics.** `backend/eval/segmentation_metrics.py`: `f1_at_overlaps`
  (F1@{10,25,50}), `average_precision`/`mean_average_precision` (mAP@tIoU), `segment_count_ratio`. tIoU alone
  is blind to a rally split in 3 or two rallies merged ? exactly what the reconciler exists to fix.

115 **Status.** Helpers + 24 unit tests landed (pure stdlib, run in the numpy/GPU-free lanes); ruff + mypy clean.
116 **Not yet wired into the default harness flow** (wiring `compare`/`run_regression.py` to report CIs + F1@k is
117 the follow-on, owner-gated). The discipline they encode is the methodology for every future cue measurement ?
118 see §13/C1?C4 for the citations and the remaining (Tier-2/3) corrections.
119
120 ### Multi-signal fusion P2: the FusionSegmenter *(in progress, 2026-06-14)*

#### 8. assistant (2026-06-14T09:49:40.824Z)

I have a good understanding of the entry style. Let me check the exact location of the "## Current track (live)" header and the first entry that follows, so I insert precisely. Let me look at lines 41-43 again and the PERSONALIZATION_PLAN.md.

#### 9. user (2026-06-14T09:49:41.253Z)

1 # Personalization Plan ? a per-user feature on a generalization-first baseline
2
3 **Status: DESIGN ? owner-adopted (hybrid); Phase-0 engineering landing.** This is the authoritative
4 plan for the personalization track. The detailed strategy analysis + the full pros/cons that led here
5 live in the working artifacts `scratch/PERSONALIZATION_PIVOT_ANALYSIS.md` (the decision-grade
6 counter-case) and `scratch/PERSONALIZATION_FEATURE_DESIGN.md` (the code-grounded design); this doc is
7 the checked-in summary + decisions + phased build plan. Peer to
8 [OWNED_MODEL_IMPLEMENTATION_PLAN.md](OWNED_MODEL_IMPLEMENTATION_PLAN.md),
9 [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md), and [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md).
10
11 ## 0. Thesis & strategic rationale (why *hybrid*, not personalization-first)
12
13 **Foundation principle (owner):** *"a tool that gets better the more you help it."* Contribution
14 (data / video / diagnostics / annotation) improves BOTH the user's own model AND the shared baseline;
15 **payoff scales with contribution.**
16
17 A directional pivot ? *personalization-first* (ship a baseline that continually self-optimizes to one
18 user's own footage) ? was researched in depth. The verdict was **HYBRID, not all-in**:
19
20 - **Adopt personalization as a flag-gated FEATURE on a generalization-first baseline.** The harness
21 genuinely makes per-user *tuning* near-free (it is already a corpus-relative A/B engine; a per-user
22 corpus is just a special case). This is real and worth shipping.
23 - **Do NOT bet the strategy on per-user *learning*.** Per-user A/B is the n?6 small-sample regime
24 *worse* (sharding data when you should pool); the differentiated parts (a learned per-user model
25 serving path, the correction?label write path, a user/tenant dimension) are largely unbuilt; a
26 naively-overfit per-user model loses the cross-user data flywheel and has no shared baseline to
27 regress against.
28 - **The owner's contribution-flywheel + free-tier-pooling closes the one real hole** the counter-case
29 found: by having (free-tier) users contribute to the *shared* baseline, the compounding cross-user
30 moat is preserved *alongside* personalization ? personalization **and** flywheel, not either/or.

Net: a generalization-first **baseline** ("batteries"), a thin **per-user tuning** layer gated by the honest small-N statistics (with a hard floor + structural-guard protection), and a **contribution loop** that feeds both.

## 1. Owner decisions (locked 2026-06-14)

1. **Identity** ? auth WILL land; *design for the north star (multi-user / cloud), implement for current use (single-user local)*. `user_id` nullable now (NULL = local install), device/install UUID for the alpha; real auth + a `users` table later.
2. **Per-user model storage** ? inline `classifier_json` in the (local) DB now, but **behind a load interface** so that post-auth it is fetched from the **cloud-hosted DB run by the service** that generates base models and orchestrates the product. The local store is one implementation of that interface.
3. **`min_n` is a PROMOTION-confidence floor, NOT a service/segmentation gate.** A brand-new user with one video is ALWAYS fully segmented by the baseline batteries ? never denied, never "give me more videos first." `min_n` only gates whether a *per-user* A/B is trusted enough to OVERRIDE a baseline cue default *for that user*; below it (or if the CI / sign test don't clear) ? keep the baseline.
   **Training-corpus policy:** *no more than needed, no less than all-if-few* ? fit on the `smallest_k_with_lift` videos (the held-out-USER learning curve), else ALL available.
4. **"Flip Gate-A to default"** = ship the validated Gate-A held-shuttle guard as the **default detection path** (today it lives in the opt-in `FusionSegmenter`, which is not the default segmenter). Changes served behavior ? **gated on a golden-set regression run + owner OK**; still open (see §5).
5. **Flywheel = backend property** (not in-product marketing). A **free tier** is likely and **free-tier videos ARE pooled** into the baseline (consent baked into the free terms; minors-exclusion still enforced) ? the clean value-exchange that feeds the owner-trained base model.
   Paid tiers get stronger privacy / opt-out.

**Pending sub-decision:** structural-guard scope ? recommended **Gate-A only** (the proven structural held-shuttle guard; its value is in failure modes a user's corpus may not yet contain), with **Gate-B** (person-occupancy / `min_players`) left *subject* to per-user A/B; **`min_n = 5`**. (Awaiting owner confirmation.)

## 2. Architecture

**Two-layer decoupling (already real).** Detection (GPU, run once): `FusionSegmenter` persists the **full superset** of per-window features into `candidate_segments.stats`; only the served handoff is gated. Decision (CPU, swappable): `experiment.Variant` + `predict()` re-score persisted features.
**Personalization == a per-user `Variant`** (cue_flags + `min_crossings`/`min_players` + optional LogReg + threshold). No segmenter rewrite ? a per-user `FusionConfig` overlay resolved

upstream.

**The single highest-leverage wiring:** `fusion_hybrid.py::_select_for_handoff` is the production decision seam. When a per-user model is pinned AND the promotion gate says ON, apply `LogisticRegression.predict_proba` over `window_stats` (the same math `experiment.predict()` runs offline). `LogisticRegression.load` is **never called in `backend/pipeline/` today** ? the serving bridge is the #1 wall.

**Per-user data model** (migrations follow the `PRAGMA user_version` ladder; current = v3):
- **v4** ? nullable `user_id` on `videos` + `candidate_segments` (NULL = local; byte-identical to today).
- **v5** ? `user_models` store: `(user_id, version, baseline_version, feature_schema_hash, fusion_config, classifier_json, promotion_evidence, n_videos_at_fit, is_active)`. Inline `classifier_json` (tenant-safe); `feature_schema_hash` is the MINOR-vs-MAJOR upgrade switch.
- **v6** ? `contribution_consent` ledger (default-deny; minors-exclusion enforced **at the pooling query**, not advisory).

## 3. Constructs (the gap list + what is built)

| # | Construct | Status |
|---|---|---|
| 1 | Per-user store + serving bridge (wire `LogisticRegression.load` into `_select_for_handoff`) | Phase 1 (the #1 wall) |
| 2 | Correction-loop ? per-user labels (`human_label` setter + confirm/reject API; recall-protecting de-bias prior) | Phase 2 (`human_label` is a read-only stub today) |
| 3 | **Per-user promotion gate** (min-N floor + structural-guard exemption) | ? **DONE ? PR #141** (`backend/eval/per_user_gate.py`) |
| 4 | Contribution flywheel (opt-in data/video/diagnostics ? shared baseline; payoff ? contribution; consent + minors-exclusion) | Phase 2 |
| 5 | User-as-annotator charter (rally-annotator: auto-generate a ~2-min high-uncertainty segment from the user's own video; guide labeling ? grounding) | Phase 3 |
| 6a | **Held-out-USER learning curve** (the falsifiable "<K videos to superior quality" criterion) | ? **DONE ? PR #142** (`backend/eval/per_user_eval.py`) |
| 6b | Per-user no-regression guard (frozen per-user CONTROL; never accept a regressing retrain/upgrade) | ? **in flight** (`feat/per-user-no-regression`) |
| 7 | Day-0 "batteries" = the validated label-free Gate-A/B fusion (?F1 +0.074, 6/6, p=0.031) shipped as the default | Phase 0, owner-gated (§5) |

## 4. Phased PR-sized build plan + status

**Phase 0 ? safety + day-0 batteries** (no schema / API / served-behavior change; pure-Python + tests):
- PR-0.1 per-user promotion gate ? ? **#141 merged**
- PR-0.2 held-out-USER learning curve ? ? **#142 merged**
- PR-0.3 per-user no-regression guard ? ? in flight
- PR-0.4 ship Gate-A as the day-0 battery ? **owner-gated** (§5; needs a golden-set regression run)
- PR-0.5 within-user LOVO measurement (the "confirm-or-kill" experiment) ? **owner-gated** (needs multi-video-per-user data on the GPU box)

**Phase 1 ? per-user tuning serving + store:** v4 migration + `set_human_label`/`set_user_id`; v5 `user_models` + CRUD; `serving.py` overlay resolver + `load_user_classifier`; **wire the bridge into `_select_for_handoff` (owner-gated ? served path)**; `/retune` + `/model` trust-feature endpoints.

```
113     **Phase 2 ? correction loop + flywheel:** `POST /api/candidates/{id}/label` (the first `human_label`
114     writer); recall-protecting de-bias prior; v6 consent ledger + minors-exclusion; **opt-in contribution
115     upload (owner-gated ? privacy surface)**; free-tier pooling.
116
117     **Phase 3 ? annotator charter + taxonomy + baseline upgrades:** active-learning task picker; rally-
118     annotator CSV ingest ? per-user labels; **coarse hard/soft shot proxy (owner-gated)**; **baseline-
119     upgrade governance (MINOR carry / MAJOR re-fit on `feature_schema_hash`) + optional federated nudge
120     (owner-gated, counsel).**
121
122     ## 5. Open decisions (owner)
123
124     1. **Structural-guard scope + `min_n` value** ? lock **Gate-A-only structural, `min_n = 5`** (recommended)?
125     2. **Gate-A default flip (PR-0.4)** ? Gate-A lives in the non-default `FusionSegmenter`, so "ship as
126        default" is a path choice (make that segmenter the default vs port the gate into the default path).
127        Recommended sequence: **run a golden-set regression first**, bring the number, *then* approve.
128     3. **Per-user `user_id` for the alpha** ? device/install UUID acceptable, or must auth land first?
129     4. **Local-install flywheel default** ? fully-local zero-pooling until consent is explicitly granted?
130
131     ## 6. Honest risks (carried from the analysis)
132
133     - **Small-N per-user eval noise** is central and *worse* per-user ? mitigated structurally: the gate
134       refuses below `min_n` and falls back to the baseline; never auto-promotes in the regime the tooling
135       exists to refuse in. PR-0.5 confirms-or-kills the core premise before any serving build-out.
136     - **Cold-start** ? the day-0 Gate-A battery (zero per-user labels) is the floor for the median user.
137     - **Privacy deferral** ? do not advertise "data never leaves your machine" until on-device ships;
138       near-term pooling carries the full cloud (DPA / consent / minors) posture.
139     - **Compute** ? heavyweight VLM personalization (on-device Qwen3-VL fine-tune) is **out of v1 scope**;
140       v1 is the cheap LogReg / threshold / cue-ON-OFF layer only.
141
142     ## 7. References
143     - `scratch/PERSONALIZATION_PIVOT_ANALYSIS.md` ? the decision-grade strategy analysis (pros/cons, harness-fit, counter-case, recommendation).
144     - `scratch/PERSONALIZATION_FEATURE_DESIGN.md` ? the code-grounded design (schema, serving bridge, APIs, critical files).
145     - Built: `backend/eval/per_user_gate.py` (#141), `backend/eval/per_user_eval.py` (#142).
146     - Harness: [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md), `backend/eval/experiment.py`, `backend/eval/eval_stats.py`.
147
```

### 10. assistant (2026-06-14T09:49:44.186Z)

I have both files. The plan confirms the exact details I need. Now let me make edit (2) first ? the PERSONALIZATION_PLAN.md structural-guard line (lines 60-63).

### 11. user (2026-06-14T09:49:48.059Z)

The file C:/Users/avidu/Projects/bhi-qiter/docs/PERSONALIZATION_PLAN.md has been updated successfully. (file state is current in your context ? no need to Read it back)

### 12. assistant (2026-06-14T09:49:51.531Z)

```
Now edit (1) ? add the new QUALITY_ITERATIONS.md entry at the TOP of the "## Current track
(live) ? newest first" section, before the existing first entry (line 43, the Fusion P3+P4
entry).
```

### 13. user (2026-06-14T09:50:04.703Z)

```
The file C:/Users/avidu/Projects/bhi-qiter/docs/QUALITY_ITERATIONS.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
```

### 14. assistant (2026-06-14T09:50:07.094Z)

```
Both edits are done. Let me verify the diff and confirm the branch state before committing. Let
me re-verify HEAD is mine and check the diff.
```

### 15. user (2026-06-14T09:50:09.355Z)

```
docs/quality-checkpoint-personalization
---HEAD---
4be16e1 docs(personalization): add PERSONALIZATION_PLAN ? hybrid per-user feature + locked owner
decisions (#144)
---vs origin/master---
---STATUS---
 M docs/PERSONALIZATION_PLAN.md
 M docs/QUALITY_ITERATIONS.md
```

### 16. assistant (2026-06-14T09:50:12.439Z)

```
Branch is mine, HEAD == origin/master (no sibling commits), and only my two files are modified.
Let me view the diff to confirm correctness before committing.
```

### 17. user (2026-06-14T09:50:13.376Z)

```
 docs/PERSONALIZATION_PLAN.md |  4 ++--
 docs/QUALITY_ITERATIONS.md   | 29 +++++++++++++++++++++++++++++
 2 files changed, 31 insertions(+), 2 deletions(-)
=====DIFF=====
diff --git a/docs/PERSONALIZATION_PLAN.md b/docs/PERSONALIZATION_PLAN.md
index 6f3f358..69e9ada 100644
--- a/docs/PERSONALIZATION_PLAN.md
+++ b/docs/PERSONALIZATION_PLAN.md
@@ -59,8 +59,8 @@ loop** that feeds both.

 **Pending sub-decision:** structural-guard scope ? recommended **Gate-A only** (the proven
structural
 held-shuttle guard; its value is in failure modes a user's corpus may not yet contain), with
**Gate-B**
-(person-occupancy / `min_players`) left *subject* to per-user A/B; **`min_n = 5`**. (Awaiting
owner
-confirmation.)
+(person-occupancy / `min_players`) left *subject* to per-user A/B; **`min_n = 5`**. **Gate-A-
only
+structural, `min_n = 5` ? owner-confirmed 2026-06-14.**

 ## 2. Architecture

diff --git a/docs/QUALITY_ITERATIONS.md b/docs/QUALITY_ITERATIONS.md
index d2d3a51..308517d 100644
--- a/docs/QUALITY_ITERATIONS.md
+++ b/docs/QUALITY_ITERATIONS.md
@@ -40,6 +40,35 @@ is the precision bug the fusion gates target.

 ## Current track (live) ? newest first
```

```
+### PERSONALIZATION Phase-0 quality checkpoint ? the eval/safety RAILS for the per-user feature
*(2026-06-14)*
+The owner adopted the **HYBRID** posture: personalization as a **flag-gated FEATURE on a
generalization-first baseline**,
+paired with a **contribution flywheel** so per-user gains and the shared baseline compound
together (full design in
+[PERSONALIZATION_PLAN.md](PERSONALIZATION_PLAN.md), merged
[#144](https://github.com/avidullu/badminton-highlight-indexer/pull/144)).
+This checkpoint landed the three **honest-eval RAILS** that keep a per-user/small-N tuning
layer from promoting noise or
+silently regressing a user:
+
+- **Per-user promotion gate ? [#141](https://github.com/avidullu/badminton-highlight-
indexer/pull/141)**
+  (`backend/eval/per_user_gate.py`): a **min-N floor** + **structural-guard exemption** ? per-
user/small-N tuning can't
+  promote noise and can't disable a structural guard (the proven Gate-A held-shuttle guard
stays on regardless of a thin
+  user corpus).
+- **Held-out-USER learning curve ? [#142](https://github.com/avidullu/badminton-highlight-
indexer/pull/142)**
+  (`backend/eval/per_user_eval.py`): turns the vague "<K videos to superior quality" claim into
a **falsifiable release
+  number** (the `smallest_k_with_lift` measured on held-out users).
+- **Per-user no-regression guard ? [#143](https://github.com/avidullu/badminton-highlight-
indexer/pull/143)**: a
+  retrain/baseline-upgrade can **never make a user worse than their incumbent** (frozen per-
user CONTROL; a regressing
+  candidate is refused).
+
+**Read HONESTLY:** these are the **eval/safety rails** for the personalization feature ?
**not** new measured
+rally-detection wins. No F1 moved this checkpoint; what landed is the discipline that lets per-
user tuning ship without
+regressing the baseline or over-claiming on n?6-per-user data.
+
+**Locked owner decisions (2026-06-14):** identity = *design for the north-star (multi-
user/cloud), implement for current
+use (single-user local)*; the inline per-user classifier sits **behind a cloud-load interface**
(local store = one impl);
+**`min_n` is a PROMOTION-confidence floor, NOT a service/segmentation gate** (a one-video user
is ALWAYS fully segmented by
+the baseline batteries); **free-tier videos ARE pooled** into the shared baseline (consent
baked into free terms,
+minors-exclusion enforced at the pooling query); and the previously-pending structural-guard
sub-decision is now LOCKED ?
+**Gate-A-only structural + `min_n = 5`**.
+
 ### Fusion P3 + P4 MEASURED on the n=6 golden corpus ? person = no lift, audio = promising (not
promotable) *(2026-06-14)*
 The GPU person-feature extraction finished all 6 golden videos (`fusion_golden.py` ?
`output/*_golden_features.json`, the
 replayable substrate); the CPU LOVO drivers then measured the lift with **honest stats (C1 ?
bootstrap 95% CI + paired sign
```

### 18. assistant (2026-06-14T09:50:15.574Z)

The diff looks correct and complete. Now let me stage the explicit paths and commit.

### 19. user (2026-06-14T09:50:17.129Z)

```
M  docs/PERSONALIZATION_PLAN.md
M  docs/QUALITY_ITERATIONS.md
```

**20. user (2026-06-14T09:50:25.650Z)**

```
[docs/quality-checkpoint-personalization ca9be06] docs(quality): personalization Phase-0
checkpoint ? eval/safety rails + locked structural-guard decision
 2 files changed, 31 insertions(+), 2 deletions(-)
```

**21. assistant (2026-06-14T09:50:27.901Z)**

Committed. Now push and create the PR.

**22. user (2026-06-14T09:50:30.689Z)**

```
remote:
remote: Create a pull request for 'docs/quality-checkpoint-personalization' on GitHub by
visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/docs/quality-
checkpoint-personalization
remote:
branch 'docs/quality-checkpoint-personalization' set up to track 'origin/docs/quality-
checkpoint-personalization'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      docs/quality-checkpoint-personalization -> docs/quality-checkpoint-
personalization
```

**23. user (2026-06-14T09:50:43.015Z)**

https://github.com/avidullu/badminton-highlight-indexer/pull/145

**24. assistant (2026-06-14T09:50:45.703Z)**

PR #145 created. Now let me check CI status to decide on merging.

**25. user (2026-06-14T09:50:47.388Z)**

```
Full test suite (offline, mocked)       pending 0       https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27495149429/job/81267713262
Mypy (lenient baseline) pending 0       https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27495149429/job/81267713284
Ruff lint       pending 0       https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27495149429/job/81267713271
Syntax sweep + import smoke (no GPU stack)      pending 0
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495149429/job/81267713260
---EXIT:8---
```

**26. assistant (2026-06-14T09:50:49.770Z)**

CI is pending. This is a docs-only change which falls within the merge agreement (docs/quality-log update). Let me wait for CI to complete using a watch loop with a reasonable timeout.

**27. user (2026-06-14T09:52:55.222Z)**

```
Refreshing checks status every 20 seconds. Press Ctrl+C to quit.

Mypy (lenient baseline) pass    29s     https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27495149429/job/81267713284
Full test suite (offline, mocked)       pending 0       https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27495149429/job/81267713262
Ruff lint       pass    8s      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27495149429/job/81267713271
Syntax sweep + import smoke (no GPU stack)      pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495149429/job/81267713260
Refreshing checks status every 20 seconds. Press Ctrl+C to quit.

Mypy (lenient baseline) pass    29s     https://github.com/avidullu/badminton-highlight-
```

indexer/actions/runs/27495149429/job/81267713284
Full test suite (offline, mocked)       pending 0       https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495149429/job/81267713262
Ruff lint       pass    8s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495149429/job/81267713271
Syntax sweep + import smoke (no GPU stack)      pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495149429/job/81267713260
Full test suite (offline, mocked)       pass    1m40s   https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495149429/job/81267713262
Mypy (lenient baseline) pass    29s     https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495149429/job/81267713284
Ruff lint       pass    8s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495149429/job/81267713271
Syntax sweep + import smoke (no GPU stack)      pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495149429/job/81267713260
Full test suite (offline, mocked)       pass    1m40s   https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495149429/job/81267713262
Mypy (lenient baseline) pass    29s     https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495149429/job/81267713284
Ruff lint       pass    8s      https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495149429/job/81267713271
Syntax sweep + import smoke (no GPU stack)      pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495149429/job/81267713260
---WATCH EXIT:0---

### 28. assistant (2026-06-14T09:52:58.397Z)

All four checks are green. This is within the merge agreement (docs/quality-log update). Let me merge with squash and delete the branch.

### 29. user (2026-06-14T09:53:02.582Z)

failed to delete local branch docs/quality-checkpoint-personalization: failed to run git: error: cannot delete branch 'docs/quality-checkpoint-personalization' used by worktree at 'C:/Users/avidu/Projects/bhi-qiter'

---EXIT:1---

### 30. assistant (2026-06-14T09:53:10.405Z)

The merge itself succeeded (remote branch deleted); only the local branch delete failed because the worktree still holds it. That's expected ? I'll clean up the worktree first, then delete the local branch. Let me verify the merge landed, then do cleanup.

### 31. user (2026-06-14T09:53:11.381Z)

{"mergeCommit":"7c6498f1d190ea042fbbfbea584c0cbce37ace6d","mergedAt":"2026-06-14T09:53:02Z","state":"MERGED"}

### 32. assistant (2026-06-14T09:53:18.040Z)

Merge confirmed (state MERGED, commit 7c6498f). Now the cleanup: remove the worktree, delete the local branch, prune.

### 33. user (2026-06-14T09:53:19.925Z)

worktree removed
Deleted branch docs/quality-checkpoint-personalization (was ca9be06).
pruned
---WORKTREE LIST---
C:/Users/avidu/Projects/badminton-highlight-indexer  4be16e1 [master]

### 34. assistant (2026-06-14T09:53:32.568Z)

All cleanup done. Note the main checkout's master still shows `4be16e1` locally (the merge happened on remote); a `git pull` in a future session will fast-forward it. The task is complete.

## Summary

**PR #145 ? MERGED (green CI).** https://github.com/avidullu/badminton-highlight-indexer/pull/145
- Squash-merge commit: `7c6498f`
- All 4 CI checks passed (full test suite 1m40s, mypy, ruff, syntax/import smoke).

### What was done

Worked in an isolated worktree (`bhi-qiter`) branched explicitly off `origin/master`. Verified HEAD had no sibling commits before committing, staged only the two explicit doc paths.

**(1) `docs/QUALITY_ITERATIONS.md`** ? Added a new entry at the TOP of "## Current track (live) ? newest first", matching the existing `### Title *(date)*` style. It captures the PERSONALIZATION Phase-0 quality checkpoint: the owner's HYBRID adoption (flag-gated per-user feature on a generalization-first baseline + contribution flywheel, design in `PERSONALIZATION_PLAN.md` / #144), and the three honest-eval RAILS ? per-user promotion gate (#141), held-out-USER learning curve (#142), per-user no-regression guard (#143). Framed honestly as eval/safety rails, NOT new measured rally-detection wins (no F1 moved). Records the locked owner decisions including `min_n` as a promotion-confidence floor (not a service gate), free-tier pooling, and the now-LOCKED Gate-A-only structural + `min_n=5`.

**(2) `docs/PERSONALIZATION_PLAN.md`** ? §1 pending sub-decision: changed the structural-guard line from "(Awaiting owner confirmation.)" to LOCKED ? "**Gate-A-only structural, `min_n = 5` ? owner-confirmed 2026-06-14.**"

### Cleanup

Worktree removed, local branch `docs/quality-checkpoint-personalization` deleted, `git worktree prune` run. Remote branch was auto-deleted on merge.

Note: the primary checkout's local `master` still points at `4be16e1` (pre-merge) since the merge landed remotely; it will fast-forward on the next `git pull` per the repo-freshness convention.